Part A

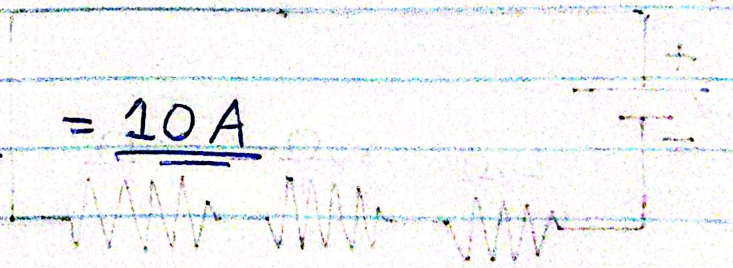
1 a. $20 + 30 + 50 = \underline{100\,\Omega}$

b. $\frac{1}{R_T} = \frac{1}{20} + \frac{1}{100} + \frac{1}{50} = \frac{5+1+2}{100} = \frac{8}{100} = \frac{1}{R_T}$ $\quad \cancel{\frac{8R_T}{8}} = \cancel{\frac{100}{8}}\ \frac{50}{\cancel{4}}\ \frac{\cancel{25}}{2}\ 12$

$= \underline{12.5\,\Omega}$

2. $\text{Current} = \frac{\text{Voltage}}{\text{Resistance}}$

a. $\frac{125}{100} = \underline{1.25A}$



b. $\frac{125}{125} = \frac{\cancel{125} \times 10}{\cancel{125}} = \underline{10A}$

3. a. $I_T \cdot I_1 = I_2 \cdot I_3 = \underline{1.25A}$

b. $\frac{125}{20} + \frac{125}{100} + \frac{125}{50} = 6.25A + 1.25 + 2.5 = \underline{10A}$

4. a. Voltage $= I \times R$

$1.25 \times 20 = 25$

$1.25 \times 30 = 37.5$

$1.25 \times 50 = 62.5$

$$\begin{array}{r} 62.5 \\ +37.5 \\ 25.0 \\ \hline 125.0 \end{array}$$

$= \underline{125V}$

b. In parallel voltage is constant

$= \underline{125V}$

5 Power dissipated

a. $P = V \times I$

$P_{20\Omega} = 25V \times 1.25A = 31.25W$

$P_{30\Omega} = 37.5V \times 1.25A = 46.875W$

$P_{50\Omega} = 62.5V \times 1.25A = 78.125W$

b. $P_{20\Omega} = 125V \times 6.25A = 781.25W$

$P_{100\Omega} = 125V \times 1.25A = 156.25W$

$P_{50\Omega} = 125V \times 2.5A = 312.5W$